

CBSE BOARD EXAMINATION 2025-26

CLASS XII – BIOLOGY (Subject Code: 044)

PREDICTIVE QUESTION BANK | Based on NCERT & Latest CBSE Pattern

Time Allowed: 3 Hours

Maximum Marks: 70

Set: A (All India)

GENERAL INSTRUCTIONS: (i) All questions are compulsory. (ii) The paper has five sections – A, B, C, D and E. (iii) Section A has 16 MCQs & Assertion–Reason questions (1 mark each). (iv) Section B has 5 Very Short Answer questions (2 marks each). (v) Section C has 7 Short Answer questions (3 marks each). (vi) Section D has 2 Case-based questions (4 marks each). (vii) Section E has 3 Long Answer questions (5 marks each). (viii) Internal choices are provided in some questions.

SECTION A – MULTIPLE CHOICE QUESTIONS (MCQs) & ASSERTION–REASON

16 × 1 = 16 Marks

■ Chapter: Ch-1: Sexual Reproduction in Flowering Plants

Q1. Which of the following is the correct sequence of events in double fertilisation in angiosperms?

[1 Mark]

- (a) Syngamy → Triple fusion → Fertilisation
- (b) Triple fusion → Syngamy → Endosperm formation
- (c) Syngamy and Triple fusion occur simultaneously
- (d) Syngamy occurs first, followed by triple fusion

✓ **Answer: (c) Syngamy and Triple fusion occur simultaneously**

■ *Examiner's Tip: Double fertilisation is unique to angiosperms — one male gamete fuses with egg (syngamy) and the other fuses with 2 polar nuclei (triple fusion) at the SAME TIME.*

Q2. Identify the INCORRECT statement about apomixis:

[1 Mark]

- (a) Apomixis is a form of asexual reproduction mimicking sexual reproduction.
- (b) Seeds produced by apomixis are genetically identical to the parent.
- (c) Apomixis always requires fertilisation to produce seeds.
- (d) Polyembryony is one form of apomixis.

✓ **Answer: (c) Apomixis always requires fertilisation to produce seeds.**

■ *Examiner's Tip: Apomixis = reproduction WITHOUT fertilisation. This is a FAVOURITE 1-mark question in boards. 'Incorrect statement' type — read all options carefully.*

■ Chapter: Ch-2: Human Reproduction

Q3. The acrosome of a sperm is derived from which organelle?

[1 Mark]

- (a) Mitochondria
- (b) Golgi complex
- (c) Endoplasmic reticulum
- (d) Centriole

✓ **Answer: (b) Golgi complex**

■ *Examiner's Tip: Acrosome = modified Golgi. Midpiece = mitochondria (energy). Tail = flagellum. This 3-part structure is extremely high-yield.*

Q4. Which hormone is responsible for the completion of meiosis-I in the primary oocyte?

[1 Mark]

- (a) FSH
- (b) LH
- (c) Progesterone
- (d) Estrogen

✓ **Answer: (b) LH (Luteinising Hormone)**

■ *Examiner's Tip: LH surge triggers ovulation AND completion of meiosis-I. The secondary oocyte arrests at meiosis-II until fertilisation — classic exam distinction.*

■ Chapter: Ch-3: Reproductive Health

Q5. Assertion (A): Amniocentesis is used for sex determination in India.

[1 Mark]

Reason (R): The technique analyses fetal chromosomes and can detect sex-linked disorders.

- (a) Both A and R are true, and R is the correct explanation of A.
- (b) Both A and R are true, but R is NOT the correct explanation of A.
- (c) A is true but R is false.
- (d) A is false but R is true.

✓ **Answer: (d) A is false but R is true.**

■ *Examiner's Tip: Amniocentesis for sex determination is BANNED in India under PCPNDT Act. However, the technique CAN detect chromosomal abnormalities — so R is true but the assertion that it is 'used for sex determination' is false legally.*

■ Chapter: Ch-4: Principles of Inheritance and Variation

Q6. In a dihybrid cross involving two independently assorting genes, the phenotypic ratio in

[1 Mark]

F₂ generation is:

- (a) 1:2:1
- (b) 3:1
- (c) 9:3:3:1
- (d) 1:1:1:1

✓ **Answer: (c) 9:3:3:1**

■ *Examiner's Tip: 9:3:3:1 is the CLASSIC dihybrid ratio — memorise it. Exception: Epistasis modifies it (9:7, 12:3:1 etc.) — likely asked in Sec C/E.*

Q7. A colour-blind woman marries a normal vision man. What will be the phenotype of their sons?

[1 Mark]

- (a) All sons will have normal vision
- (b) All sons will be colour blind
- (c) 50% sons colour blind
- (d) All daughters colour blind

✓ **Answer: (b) All sons will be colour blind**

■ *Examiner's Tip: Colour blind woman = $X^c X^c$. Normal man = $X^N Y$. Sons get X^c from mother → all colour blind. This pedigree logic is ALWAYS in the paper.*

■ Chapter: Ch-5: Molecular Basis of Inheritance

Q8. Which enzyme is responsible for the removal of RNA primer during DNA replication in prokaryotes?

[1 Mark]

- (a) DNA Polymerase I
- (b) DNA Polymerase III
- (c) Primase
- (d) Ligase

✓ **Answer: (a) DNA Polymerase I**

■ *Examiner's Tip: DNA Pol III = main replication enzyme. DNA Pol I = removes RNA primer and fills gap. DNA Ligase = joins Okazaki fragments. This distinction is asked every year.*

Q9. Which of the following is NOT a feature of the genetic code?

[1 Mark]

- (a) Degenerate
- (b) Overlapping
- (c) Universal
- (d) Non-ambiguous

✓ **Answer: (b) Overlapping**

■ *Examiner's Tip: The genetic code is NON-OVERLAPPING. Other features: triplet, degenerate, commaless, non-ambiguous, nearly universal. Expect this in MCQ every board cycle.*

■ Chapter: Ch-6: Evolution

Q10. The concept of 'survival of the fittest' was coined by:

[1 Mark]

- (a) Charles Darwin
- (b) Alfred Wallace
- (c) Herbert Spencer
- (d) Jean-Baptiste Lamarck

✓ **Answer: (c) Herbert Spencer**

■ *Examiner's Tip: Darwin proposed 'natural selection'. Spencer coined 'survival of the fittest'. Wallace co-discovered natural selection. These distinctions are 1-mark gold.*

■ Chapter: Ch-7: Human Health and Disease

Q11. Assertion (A): Cancer cells do not exhibit contact inhibition.

[1 Mark]

Reason (R): Normal cells stop dividing when they come in contact with other cells.

- (a) Both A and R are true, and R is the correct explanation of A.
- (b) Both A and R are true, but R is NOT the correct explanation of A.
- (c) A is true but R is false.
- (d) A is false but R is true.

✓ **Answer: (a) Both A and R are true, and R is the correct explanation of A.**

■ *Examiner's Tip: Contact inhibition is the key difference between normal and cancerous cells. Perfect Assertion-Reason answer: A is true AND R correctly explains A.*

■ Chapter: Ch-8: Microbes in Human Welfare

Q12. Which microorganism is used in the production of streptokinase, a 'clot buster' drug?

[1 Mark]

- (a) *Aspergillus niger*
- (b) *Streptococcus*
- (c) *Trichoderma*
- (d) *Monascus purpureus*

✓ **Answer: (b) Streptococcus**

■ *Examiner's Tip: Memory hook: STREPtokinase → STREPtococcus. Aspergillus = citric acid. Monascus = blood cholesterol lowering. Trichoderma = biocontrol.*

■ Chapter: Ch-9: Biotechnology – Principles and Processes

Q13. The correct order of steps in recombinant DNA technology is:

[1 Mark]

- (a) Isolation → Amplification → Cloning → Expression
- (b) Cloning → Isolation → Amplification → Expression
- (c) Amplification → Isolation → Cloning → Expression
- (d) Isolation → Cloning → Amplification → Expression

✓ **Answer: (a) Isolation → Amplification → Cloning → Expression**

■ *Examiner's Tip: Sequence: Identify gene → Isolate DNA → Amplify (PCR) → Insert into vector (Cloning) → Transform host → Expression. Know this flow cold.*

■ Chapter: Ch-10: Biotechnology and its Applications

Q14. Bt toxin produced by *Bacillus thuringiensis* is inactive in the bacterium itself because:

[1 Mark]

- (a) It is present in an inactive pro-toxin form
- (b) The bacterium has a thick cell wall
- (c) The pH inside the bacterium is acidic
- (d) The bacterium lacks ribosomes to translate it

✓ **Answer: (a) It is present in an inactive pro-toxin form**

■ *Examiner's Tip: Bt toxin = protoxin (inactive crystal protein). Gets activated by alkaline pH of insect midgut → lyses midgut cells. This mechanism is asked almost every year.*

■ Chapter: Ch-11: Organisms and Populations

Q15. Which of the following is an example of intraspecific competition?

[1 Mark]

- (a) Lion and hyena competing for the same prey
- (b) Two tigers competing for the same territory
- (c) Mycorrhiza on plant roots
- (d) Predation by a hawk on a mouse

✓ **Answer: (b) Two tigers competing for the same territory**

■ *Examiner's Tip: INTRA = same species. INTER = different species. Competition is most intense within same species (intraspecific) — Gause's Competitive Exclusion Principle applies to interspecific.*

■ Chapter: Ch-16: Biodiversity and Conservation

Q16. According to IUCN, species whose population has declined by more than 80% over the last 10 years are classified as:

[1 Mark]

- (a) Vulnerable
- (b) Endangered
- (c) Critically Endangered
- (d) Extinct in the Wild

✓ **Answer: (c) Critically Endangered**

■ *Examiner's Tip: IUCN thresholds: CR = >80% decline; Endangered = >50%; Vulnerable = >30%. Critically Endangered = highest risk before extinction. Boards love this specific data.*

SECTION B – VERY SHORT ANSWER QUESTIONS

5 × 2 = 10 Marks

Q17. Define 'senescence' of seeds. What is the role of 'seed dormancy' in plant survival?

[2 Marks]

✓ **Answer: Senescence = deterioration of seed viability over time due to metabolic exhaustion. Seed dormancy allows seeds to withstand unfavourable conditions and germinate only when environmental conditions (temperature, moisture, light) are suitable — ensuring species survival.**

■ *Examiner's Tip: 2-mark answer = definition + one application/significance. Never leave a 2-marker without both parts.*

Q18. Differentiate between oogenesis and spermatogenesis on the basis of: (i) Number of functional gametes produced per meiosis (ii) Site of occurrence in humans. **[2 Marks]**

✓ **Answer:** (i) *Spermatogenesis → 4 functional sperms per meiosis; Oogenesis → 1 functional ovum + 3 polar bodies (non-functional).* (ii) *Spermatogenesis: Seminiferous tubules of testes; Oogenesis: Ovarian follicles of ovary.*

■ *Examiner's Tip: Tabular difference questions — always address BOTH points asked. Missing one point = losing 1 mark.*

Q19. What is a 'linkage group'? How does linkage affect Mendel's law of independent assortment? **OR** **[2 Marks]**

Explain codominance with the help of ABO blood group system.

✓ **Answer:** *Linkage Group: Genes located on the same chromosome form a linkage group and tend to be inherited together. Linkage violates the Law of Independent Assortment because linked genes do NOT assort independently — they move together during gamete formation. [OR] Codominance: Both alleles express equally in heterozygote. In ABO blood groups, I^AI^B genotype shows BOTH A and B antigens → AB blood group, demonstrating codominance.*

■ *Examiner's Tip: Internal choice — always attempt the one you know better. Both options are equally likely in the actual paper.*

Q20. Name the parts labelled (i) and (ii) in the diagram of a transcription unit and state the function of each. **[2 Marks]**

[Note: Diagram shows Promoter, Structural gene, Terminator]

✓ **Answer:** (i) *Promoter: Binds RNA polymerase; initiates transcription — present at 5' (upstream) end of template strand.* (ii) *Terminator: Signals the end of transcription; RNA polymerase dissociates from DNA template.*

■ *Examiner's Tip: Diagram-based questions in Section B are common. Even without the diagram, know: Promoter (5') → Structural gene → Terminator (3') on template strand.*

Q21. What is 'eutrophication'? Name any TWO effects of cultural eutrophication on aquatic ecosystems. **[2 Marks]**

✓ **Answer:** *Eutrophication: Excessive enrichment of a water body with nutrients (nitrates, phosphates) leading to algal bloom. Effects: (i) Depletion of dissolved oxygen → death of aquatic animals (BOD increases). (ii) Loss of aquatic biodiversity due to algal bloom blocking sunlight.*

■ *Examiner's Tip: Eutrophication = 'cultural' when caused by human activities (sewage, fertiliser runoff). Natural eutrophication is a slow process. Know both types.*

SECTION C – SHORT ANSWER QUESTIONS

7 × 3 = 21 Marks

Q22. Explain the structure of a mature pollen grain with a neat labelled diagram. Why is the outer wall (exine) of pollen grain hard? **[3 Marks]**

✓ **Answer:** *Structure: Mature pollen grain is 2-celled — Vegetative cell (larger, with food reserve) and Generative cell (smaller, divides to form 2 male gametes). Wall: Two-layered — Exine (outer, hard, with sporopollenin — most resistant biological material, withstands extreme temperatures, acids, alkalis) and Intine (inner, cellulose + pectin). Significance of Exine: Sporopollenin makes pollen grains resistant to physical and biological degradation, ensuring dispersal over long distances.*

■ *Examiner's Tip: Draw and label: Exine, Intine, Vegetative nucleus, Generative cell, Germ pore. NCERT diagram is sufficient. Diagram = 1 mark.*

Q23. Describe the events of the 'follicular phase' of the menstrual cycle. Name the hormones involved and their sources. **[3 Marks]**

✓ **Answer: Follicular Phase (Day 1-13):** (i) FSH (from anterior pituitary) stimulates primary follicle to develop into Graafian follicle. (ii) Developing follicle secretes estrogen → causes proliferation of endometrium. (iii) Rising estrogen exerts negative feedback on FSH and positive feedback → triggers LH surge at day 13. Hormones: FSH (anterior pituitary) → stimulates follicle; Estrogen (Graafian follicle) → proliferates endometrium; LH (anterior pituitary) → triggers ovulation.

■ **Examiner's Tip:** Menstrual cycle = 28 days. Follicular (1-13), Ovulation (14), Luteal (15-28). All 4 hormones (FSH, LH, Estrogen, Progesterone) and their sources must be memorised.

Q24. What are 'co-dominance' and 'incomplete dominance'? Distinguish between them giving one example of each. **OR** **[3 Marks]**

Explain Chromosomal Theory of Inheritance. How did Morgan's experiments provide evidence for this theory?

✓ **Answer: Co-dominance:** Both alleles express simultaneously and equally in heterozygote. Example: ABO blood group — $I^A I^B \rightarrow AB$ group (both A and B antigens expressed). **Incomplete Dominance:** Neither allele is completely dominant; F1 shows intermediate phenotype. Example: Snapdragon (*Antirrhinum*) — Red (RR) × White (rr) → Pink (Rr) in F1. **Distinction:** In codominance both phenotypes appear; in incomplete dominance an intermediate/blended phenotype appears. **[OR] Chromosomal Theory (Sutton & Boveri, 1902):** Genes are located on chromosomes; behaviour of chromosomes during meiosis parallels Mendel's laws. **Morgan's evidence:** Worked on *Drosophila*; showed that genes on same chromosome are linked and do not follow independent assortment; calculated recombination frequency to map genes on chromosomes.

■ **Examiner's Tip:** 3-mark answer: Definition + Example + Distinction. For OR: Theory + Morgan's contribution. Both options are from high-frequency chapters.

Q25. Draw a neat labelled diagram of the transcription unit and describe the process of transcription in prokaryotes. **[3 Marks]**

✓ **Answer: Transcription Unit:** Promoter → Structural gene (template strand 3'→5') → Terminator. **Process in Prokaryotes:** (i) **Initiation:** RNA polymerase binds to promoter (sigma factor assists). (ii) **Elongation:** RNA polymerase moves 3'→5' along template strand, synthesises mRNA 5'→3' using ribonucleotides. (iii) **Termination:** At terminator sequence, RNA polymerase falls off; mRNA released. **Note:** In prokaryotes, transcription and translation occur simultaneously (coupled).

■ **Examiner's Tip:** Diagram: Draw template strand with labels: 3'→5' direction, coding strand, mRNA direction 5'→3', RNA polymerase. This 3-mark question has appeared in 2020, 2022, 2024.

Q26. Explain the role of microbes in the biological treatment of sewage. What are the two stages involved? **[3 Marks]**

✓ **Answer: Sewage Treatment using Microbes:** **Stage 1 – Primary Treatment (Physical):** Floating debris removed by sequential filtration; sedimentation removes grit. **Stage 2 – Secondary Treatment (Biological):** (i) **Aeration:** Sewage mixed with 'activated sludge' (bacteria-rich) in aeration tank → aerobic microbes digest organic matter → BOD reduced significantly. (ii) **Anaerobic Sludge Digestion:** Remaining sludge digested by anaerobic bacteria in sludge digester → produces biogas (methane + CO₂ + H₂S). **Significance:** Reduces BOD to safe levels before releasing into water bodies.

■ **Examiner's Tip:** BOD = Biochemical Oxygen Demand. Higher BOD = more pollution. Boards always ask: 'How does BOD indicate pollution?' Include this in longer answers.

Q27. What is the significance of 'genetic engineering' in agriculture? Explain with TWO examples of genetically modified crops.

[3 Marks]

✓ **Answer: Significance: Allows introduction of specific genes (disease resistance, pest resistance, nutritional improvement) into crop plants. Examples: (i) Bt Cotton: Gene from *Bacillus thuringiensis* (cry genes) inserted → produces Bt toxin that kills bollworm (*Helicoverpa armigera*). Reduces pesticide use. (ii) Golden Rice: Beta-carotene biosynthesis genes inserted into rice → produces Vitamin A precursor → addresses Vitamin A deficiency. Other examples acceptable: Flavr Savr tomato (delayed ripening), Bt Brinjal (insect resistance).**

■ **Examiner's Tip:** Bt Cotton and Golden Rice are the two MOST asked GM crop examples. Always give: crop name + gene source + benefit.

Q28. Explain the concept of 'keystone species'. How does the removal of a keystone species affect an ecosystem? Give one example.

[3 Marks]

✓ **Answer: Keystone Species: A species that has a disproportionately large effect on its ecosystem relative to its abundance. Effect of Removal: Removal of keystone species causes a dramatic shift in ecosystem structure — other species may increase or collapse, leading to loss of biodiversity and altered food webs. Example: Sea otter (*Enhydra lutris*) in Pacific kelp forests — sea otters control sea urchin populations; without otters, urchins overgraze kelp → destruction of kelp forest ecosystem. Indian example: Tiger as keystone predator — controls deer/herbivore populations, maintaining forest vegetation.**

■ **Examiner's Tip:** Keystone species = concept from ecology chapters (Ch 13/15). Increasingly asked in recent CBSE papers. Give a local (Indian) example if possible for extra precision.

SECTION D – CASE-BASED QUESTIONS

2 × 4 = 8 Marks

Q29. Read the following passage carefully and answer the questions that follow:

Meena is a 28-year-old woman diagnosed with complete blockage of both fallopian tubes. Her husband has a normal sperm count. After counselling, their doctor suggested they opt for In Vitro Fertilisation (IVF). The procedure involves retrieving eggs after hormonal stimulation, fertilising them in a culture dish, and transferring the resulting embryo into Meena's uterus. The doctor mentioned that if embryo transfer is not possible, a surrogate mother could carry the embryo. Additionally, GIFT and ICSI were mentioned as alternatives.

(i) What does IVF stand for? Where does fertilisation occur in this procedure? [1]

Answer: IVF = In Vitro Fertilisation. Fertilisation occurs outside the body in a culture dish (in vitro = in glass/outside the body).

(ii) Define GIFT. In what condition is GIFT preferred over IVF? [1]

Answer: GIFT = Gamete Intra Fallopian Transfer. Gametes (egg and sperm) are transferred into the fallopian tube. GIFT is preferred when at least ONE fallopian tube is functional (unlike Meena's case).

(iii) What is ICSI? How does it differ from IVF? [1]

Answer: ICSI = Intra Cytoplasmic Sperm Injection. A single sperm is directly injected into the egg cytoplasm. Difference: IVF = sperm and egg mixed in culture; ICSI = single sperm mechanically injected. ICSI used in cases of low sperm count/poor motility.

(iv) What ethical/legal consideration is involved in surrogacy in India? Mention one. [1]

Answer: The Surrogacy (Regulation) Act 2021 in India allows only 'altruistic surrogacy' (by a close relative) — commercial surrogacy is banned. Ethical concern: Exploitation of surrogate mothers for monetary gain.

■ **Tip:** ART (Assisted Reproductive Technologies) — IVF, GIFT, ZIFT, IUT, ICSI — Learn full forms + conditions for each.

Q30. Read the following passage carefully and answer the questions that follow:

A research team studying a forest ecosystem measured the energy content at each trophic level. They found: Producers = 1,000,000 kcal; Primary consumers = 100,000 kcal; Secondary consumers = 10,000 kcal; Tertiary consumers = 1,000 kcal. They noted that the forest had several food chains and food webs. The team also observed that decomposers played a crucial role in nutrient cycling but were not included in the trophic pyramid. The team studied both grazing food chains and detritus food chains in the forest.

(i) Calculate the percentage of energy transferred from producers to primary consumers. [1]

Answer: *Energy transfer = $(100,000 / 1,000,000) \times 100 = 10\%$. This follows the 10% Law (Lindemann's Law, 1942).*

(ii) Why are food chains generally limited to 3-4 trophic levels? [1]

Answer: *Due to the 10% Law — only 10% energy passes to the next trophic level. By the 4th trophic level, available energy is too low to support another consumer level.*

(iii) Differentiate between grazing food chain and detritus food chain. [1]

Answer: *Grazing Food Chain (GFC): Starts with living plants/producers; energy flows through herbivores to carnivores. Detritus Food Chain (DFC): Starts with dead organic matter (detritus); energy flows through decomposers/detrivores. DFC is more important in forest ecosystems.*

(iv) Name the biogeochemical cycle in which both gaseous and sedimentary phases are present. [1]

Answer: *The Nitrogen cycle and Carbon cycle have both phases. However, the Phosphorus cycle is purely sedimentary (no gaseous phase). Most common answer expected: Nitrogen cycle.*

■ *Tip: Ecosystem chapter — Energy flow, Productivity, Decomposition, Nutrient cycling, and Ecosystem services are ALL high-yield. Lindemann's 10% Law is asked almost every year.*

SECTION E – LONG ANSWER QUESTIONS

3 × 5 = 15 Marks

Q31. With the help of a neat labelled diagram, describe the process of DNA replication in E.

[5 Marks]

coli (prokaryotes). Mention the names and roles of all enzymes involved. **OR**

Describe the lac operon model proposed by Jacob and Monod. What happens to the operon when lactose is present?

✓ **Answer:** *DNA Replication (Semi-conservative, proved by Meselson-Stahl experiment): 1. Initiation: Helicase unwinds double helix at origin of replication (oriC); SSB proteins stabilise single strands; Topoisomerase relieves torsional strain. 2. Primer Formation: Primase synthesises short RNA primer (5'→3') on both strands. 3. Elongation: DNA Pol III adds dNTPs in 5'→3' direction — continuous on leading strand, discontinuous (Okazaki fragments) on lagging strand. 4. Primer Removal: DNA Pol I removes RNA primer, fills gap. 5. Ligation: DNA Ligase joins Okazaki fragments. Enzyme Summary: Helicase → unwinds; Primase → primer; DNA Pol III → synthesis; DNA Pol I → removes primer; Ligase → joins; Topoisomerase → removes supercoiling. [OR] LAC OPERON: Structural genes: lacZ (beta-galactosidase), lacY (permease), lacA (transacetylase). When lactose ABSENT: Repressor protein (from regulator gene i) binds to operator → blocks transcription. When lactose PRESENT: Lactose → Allolactose (inducer) → binds repressor → repressor changes shape → cannot bind operator → RNA polymerase binds promoter → transcription occurs → enzymes produced → lactose metabolised. This is an INDUCIBLE operon.*

■ *Examiner's Tip: 5-mark answer = must have diagram + all enzymes labelled. The OR (lac operon) is equally important. Both appear in almost every board paper. Memorise: Inducible vs Repressible operons.*

Q32. Explain the Darwinian theory of evolution. What are its drawbacks? How does Neo-Darwinism (Modern Synthetic Theory) overcome these limitations? **OR**
Describe the origin of life on Earth according to the Oparin-Haldane hypothesis. What was the significance of Miller and Urey's experiment?

[5 Marks]

✓ **Answer: Darwinian Theory (Natural Selection):** Postulates: (i) Overproduction — organisms produce more offspring than can survive. (ii) Struggle for existence — competition for limited resources. (iii) Variation — heritable variations exist. (iv) Natural Selection — individuals with favourable variations survive ('Survival of the fittest' — Spencer). (v) Speciation — selected variations accumulate over generations → new species. Drawbacks: (i) Did not explain source/mechanism of variations. (ii) Did not know about genetics/heredity. (iii) Could not explain evolution of neutral traits. Neo-Darwinism (Modern Synthetic Theory) — adds: Gene mutation, Chromosomal aberration, Genetic recombination, Genetic drift, Gene flow, Natural selection acts on genetic variation. Bridges Darwin + Genetics + Population genetics. [OR] Oparin-Haldane Hypothesis: Early Earth atmosphere = reducing (H₂, NH₃, CH₄, H₂O vapour, no O₂). Energy from lightning and UV radiation caused abiotic synthesis of organic molecules → 'Primordial soup'. Molecules aggregated → coacervates → proto-cells → first living cells (heterotrophs). Miller-Urey Experiment (1953): Simulated early Earth conditions → produced amino acids from inorganic molecules. Significance: First experimental evidence that organic molecules can form abiotically.

■ **Examiner's Tip:** Evolution is a 5-mark staple every year. Darwinism vs Lamarckism vs Neo-Darwinism comparisons are common. Miller-Urey setup diagram (flask with electrodes, condenser, water) may be asked.

Q33. What is biodiversity? Explain the three levels of biodiversity. Discuss any FOUR causes of biodiversity loss and suggest TWO conservation measures. **OR**
Describe the important steps involved in recombinant DNA technology (rDNA technology). Name the tools used at each step.

[5 Marks]

✓ **Answer: Biodiversity:** The variety of life on Earth at all its levels — genetic, species and ecosystem. 3 Levels: (i) Genetic diversity — variation within a species (e.g., >50,000 rice varieties in India). (ii) Species diversity — variety of species in an area (measured by species richness). (iii) Ecosystem diversity — variety of biomes/ecosystems (deserts, rainforests, wetlands). Causes of Loss (Evil Quartet): (i) Habitat loss and fragmentation — deforestation, urbanisation (most important cause). (ii) Overexploitation — hunting, poaching, overfishing. (iii) Alien species invasion — invasive species displace native species. (iv) Co-extinctions — when one species (e.g., host plant) goes extinct, dependent species also perish. Conservation Measures: (i) In situ conservation — protect species in their natural habitat (national parks, biosphere reserves, wildlife sanctuaries). (ii) Ex situ conservation — preserve outside natural habitat (zoos, botanical gardens, seed banks, cryopreservation). [OR] rDNA Technology Steps: 1. Isolation of DNA — CTAB/SDS method to lyse cells and extract genomic DNA. 2. Cutting of DNA — Restriction endonucleases (restriction enzymes) cut DNA at specific palindromic sequences → sticky ends. 3. Amplification — PCR (Polymerase Chain Reaction) amplifies specific DNA fragment using Taq polymerase. 4. Ligation — DNA ligase joins insert DNA into vector (plasmid/bacteriophage) → recombinant DNA. 5. Transformation — Recombinant vector introduced into host cell (E. coli) by electroporation/heat shock. 6. Screening/Selection — Selectable markers (antibiotic resistance) identify transformed cells. 7. Expression — Host cell expresses foreign gene → desired protein produced.

■ **Examiner's Tip:** Biodiversity 5-marker: Always include 'Evil Quartet' (4 causes) + in situ vs ex situ conservation. The OR (rDNA tech) requires ALL 7 steps with enzyme names. Both are perennial favourites.

■ HIGH-PRIORITY TOPICS FOR CBSE 2025-26 BIOLOGY BOARD EXAM

CHAPTER	TOPIC	EXPECTED IN
Ch 2 – Human Reproduction	Spermatogenesis & Oogenesis, Menstrual cycle, Embryo development	Sec C / E
Ch 4 – Inheritance	Dihybrid cross, Sex-linked inheritance, Pedigree analysis	Sec A / C / E
Ch 5 – Mol. Basis	DNA replication, Transcription, Translation, Lac operon	Sec A / C / E
Ch 6 – Evolution	Darwin vs Lamarck, Modern synthesis, Hardy-Weinberg	Sec A / C / E

Ch 7 – Health	Immunity (Active/Passive), Cancer, AIDS, Drugs	Sec A / B / C
Ch 9 – Biotech	rDNA tools, PCR, Electrophoresis, Cloning vectors	Sec A / C / E
Ch 10 – Biotech Appln	Bt cotton, Golden rice, Gene therapy, GM organisms	Sec A / C
Ch 13 – Organisms & Pop	Population interactions (table), Growth curves, Adaptations	Sec C / E
Ch 14 – Ecosystem	Energy flow, Lindemann's 10%, Nutrient cycles, Succession	Sec D / E
Ch 16 – Biodiversity	IUCN categories, Evil Quartet, In situ/Ex situ conservation	Sec A / C / E

■ All answers are based on NCERT Class 12 Biology Textbook (latest edition) and official CBSE Curriculum 2025-26. ■
 Marking scheme follows official CBSE guidelines. ■ Internal choices (OR) match the format used in CBSE sample papers and
 past 10-year question papers.